Reflection: Working at a bit level: Arnav Krishnan

I learned a lot about working at the bit level. It really helped me understand how systems that we use work at a very basic level (bitwise operations: AND, OR, XOR, etc.; pull-up resistors; input vs. output; etc). It felt like working much closer to the hardware, which helps me understand how the hardware and software interact. Being able to use these techniques to program embedded systems was a big shift from higher-level programming I have done in the past, starting with python, and then java, and finally c (getting closer and closer to assembly).

I'm still wondering how these low-level operations are "scaled up" to more complex systems, such as modern operating systems. I also wonder how errors in binary expression, which we dealt with a lot in our "Exceptions" branch of IntegerConvert, are handled in the real world (and what happens when they are not handled correctly).